When America Plugs In: A Visual Analysis of Weather and EV Charging in U.S. Metros

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1. Introduction

Electric vehicle (EV) charging is emerging as a non-trivial component of urban electricity demand as EV adoption accelerates across major U.S. metropolitan areas. Unlike many traditional electricity loads, EV charging behavior is inherently flexible in both timing and location. Drivers can adjust when and where they charge, as well as the frequency and intensity of charging, depending on prevailing conditions and travel needs. This flexibility creates the possibility that short-run environmental conditions may systematically shape charging behavior across space and time. Thus, this study will focus on the impacts of daily weather variation on charging behaviors.

Weather condition affects EV charging activity through two distinct channels. First, temperature directly alters battery performance and vehicle operating efficiency. Cold temperatures reduce battery efficiency and driving range through increased battery energy consumption and additional heating demand. High temperatures similarly increase energy consumption through air-conditioning loads and battery management systems. These mechanisms enhance the effective energy required per mile travel and may therefore increase charging frequency, charging duration, and energy delivered per session.

Second, adverse weather conditions may influence driver behavior more broadly by increasing the inconvenient cost of outdoor mobility. Rain, snow, or extreme weather events can alter normal commuting patterns, suppress non-necessary trips, and shift both the timing and location of charging decisions. Therefore, weather affects EV charging not only through the physical performance of batteries, but also through behavioral responses that determine whether trips occur at all.

These responses may additionally depend on charging technology and charging context. Level 2 (L2) charging systems deliver alternating-current power that must be converted within the vehicle, resulting in slower charging speeds and longer staying times. Direct current fast charging (DCFC) bypasses this conversion process and delivers power directly to the battery, providing faster charging and serving drivers with more immediate needs of traveling. Weather-related charging responses may therefore differ between L2 and DCFC environments. Similarly, workplace charging, retail charging, and other public charging venues may exhibit different behavioral sensitivity to environmental conditions.

Motivated by the mechanisms and the demand, this study examines how daily weather conditions shape EV charging behavior across 29 U.S. metropolitan areas between 2020 and 2022 using the U.S. Department of Energy (DOE) EV Watts public charging-session database merged with metro-level weather observations. We focus on two central research questions: i), how do temperature, precipitation, and snowfall affect EV charging outcomes including charging frequency, session duration, and energy consumption? ii), to what extent do these relationships vary across different contexts, including climate zones, charging technologies, and charging venue types?

Our analysis combines descriptive evidence with fixed-effect-adjusted empirical analysis to distinguish between those two channels of EV charging demand. These results of this study provide evidence on how both battery physics and driver behavior jointly shape weather sensitivity in emerging EV charging networks across the United States.

2. Data

2.1 EV Charging Session Data

2.1.1 Data source

EV charging session data were obtained from the EV Watts Public Database, hosted on the Livewire Data Platform and maintained by the U.S. DOE. The dataset provides information on plug-in electric vehicle charging sessions and electric vehicle supply equipment (EVSE) from 2019 to 2022, covering public charging infrastructure across the United States metropolitans.

The EV Watts documentation defines the key infrastructure units as follows: an EVSE is a single unit of electric vehicle service equipment reporting data from a number of connectors on that unit. The number of ports is a count of connectors that can be simultaneously activated while connectors are described as individual cord sets that could be plugged into a vehicle.

2.1.2 Variable list

The EV Watts session table contains 19 variables **(Table 1)**.

Table 1: EV Watts session-level variables.

Variable	Data Type	Description
session_id	PK (int)	Unique identifier for each charging session
evse_id	FK (int)	Identifier for the charging station unit
connector_id	FK (int)	Identifier for the specific connector used
metro_area	string	Metropolitan area where the EVSE resides (e.g., "Los Angeles", "Undesignated")
land_use	string	Geographic classification: metro, nonmetro, or undesignated
region	string	U.S. Census division (Northeast, Midwest, South, West)
num_ports	integer	Number of ports capable of simultaneous charging
charge_level	string	Charging rate: L2 (Level 2) or DCFC (DC Fast Charging)
venue	string	Type of location (e.g., Retail, Hotel, Multi-Unit Dwelling, Fleet)
pricing	string	Payment status: Paid, Free, or Undesignated
connector_type	string	Hardware type: CHAdeMO, Combo (CCS), or J1772
power_kw	float	Power rating of the connector (kW)
start_datetime	timestamp	Date and time the session began
end_datetime	timestamp	Date and time the session ended
total_duration	float	Total elapsed session time (hours)
charge_duration	float	Time energy was actively flowing to the vehicle (hours)
energy_kwh	float	Total energy delivered during the session (kWh)
start_soc	percent	Battery state of charge at session start (DCFC only)
end_soc	percent	Battery state of charge at session end (DCFC only)
flag_id	FK (int)	Data quality flag; 0 = clean record using binary logic to identify data quality issues

2.1.3 Cleaning pipeline

The raw session data were cleaned in ten sequential steps. Each step is reported with its rationale so the magnitude of each filter is transparent.

2.1.3.1 Load raw session data

The raw file evwatts.public.session.csv was loaded into a pandas DataFrame. The session table contains core transaction records (one row per charging event) without station metadata.

Note: start_soc and end_soc were entirely NaN across all session records in this release and were not used in any analysis.

2.1.3.2 Parse datetimes and extract date

The start_datetime and end_datetime columns were cast to pandas datetime format. A new column date was extracted from start_datetime (calendar date only) to serve as the merge key with the daily weather dataset.

2.1.3.3 Load EVSE metadata

The station-level file evwatts.public.evse.csv was loaded separately. This table contains geographic and infrastructure attributes that are absent from the session table and are required for the weather merge.

2.1.3.4 EVSE ID overlap check

Before merging, an overlap check on evse_id was performed. Table 2 reports the result.

Table 2: EVSE identifier overlap between session and EVSE tables.

Metric	Count
Unique evse_id in session data	38,458
Unique evse_id in EVSE data	36,926
Matching evse_ids	36,894
In session but not in EVSE	1,564
In EVSE but not in session	32
Match rate	≈95.9%

2.1.3.5 Merge session and EVSE tables

The session and EVSE tables were joined using an inner join on evse_id. A left join was rejected because metro area is essential for the downstream weather merge; sessions without a matching EVSE record cannot be assigned a geographic location. The inner join discarded 1,564 unmatched session records (≈4.1% of total sessions).

2.1.3.6 Drop undesignated metro area

Rows where the metro_area was labeled 'Undesignated' were removed. These records have no geographic assignment and cannot be matched to a weather series.

2.1.3.7 Drop flagged sessions

All rows with flag_id was greater than 0 were discarded. EV Watts documentation describes flag_id as abinary-logic error code identifying known data quality issues; a value of 0 indicates a clean record.

2.1.3.8 Energy outlier analysis

Descriptive statistics and a histogram were generated to identify abnormal values. Table 3 reports the key statistics.

Table 3: Descriptive statistics for energy_kwh before cleaning.

Statistic	Value
Minimum	0.00 kWh
Maximum	936.21 kWh
Mean	16.74 kWh
Median	10.83 kWh
Std. dev.	19.94 kWh
90th percentile	38.89 kWh
95th percentile	49.08 kWh
99th percentile	69.68 kWh
99.9th percentile	246.55 kWh
Sessions => 150 kWh	26,121
Sessions = 0 kWh	13,397

Three anomalies were identified: (i) a maximum of 936 kWh is physically impossible for a single charging session; (ii) 26,121 sessions exceeded 150 kWh, warranting charge-level-specific review; (iii) 13,397 sessions recorded 0 kWh, indicating failed or ghost sessions with no energy delivered. Besides, the large gap between the 99th percentile (69 kWh) and the 99.9th percentile (246 kWh) confirms a heavy right tail requiring targeted treatment.

2.1.3.9 Energy cleaning by charge level

Three actions were taken based on the outlier analysis and charge level breakdown:

1. Drop zero-kWh sessions. 13,397 sessions with energy_kwh == 0 were removed; these represent failed or phantom records with no energy delivered.
2. Drop implausible L2 sessions. Level 2 chargers are rated at a maximum of approximately 19 kW. It is implausible to achieve 150 kWh in a single session. Thus, 43 outliers were removed.
3. Cap extreme direct current fast chargers (DCFC) outliers. DCFC sessions above the 99.9th percentile within the DCFC sub-population were removed. DC fast chargers can realistically deliver 50–350 kW, so the 26,078 DCFC sessions exceeding 150 kWh were largely retained; only the extreme tail was discarded.

2.1.3.10 Export to Parquet

The cleaned session DataFrame was exported to data/processed/EV_merged.parquet using PyArrow. Datetime columns were converted to strings prior to export to resolve a known compatibility issue with Python 3.13.

2.1.4 Final EV dataset summary

Table 4 reports key properties of the cleaned EV dataset.

Table 4: EV Watts cleaned dataset summary.

Property / Filter	**Value**
Unique metro areas	29
Date range	March 2020 – December 2022
Key merge columns	date + metro_area
Dropped: unmatched EVSE	1,564 sessions
Dropped: undesignated metro	(see cleaning log)
Dropped: flag_id > 0	(see cleaning log)
Dropped: zero-kWh sessions	13,397 sessions
Dropped: implausible L2 > 150 kWh	43 sessions

Dropped: extreme DCFC outliers	above 99.9th pct within DCFC

2.2 Weather Data

2.2.1 Data source: Open-Meteo Historical Weather API

Weather data were obtained from the Open-Meteo Historical Weather Archive, a free, open-source meteorological data service available at https://open-meteo.com/en/docs/historical-weather-api. Open-Meteo provides retrospective daily weather data derived from ERA5 reanalysis (European Centre for Medium-Range Weather Forecasts, ECMWF) blended with high-resolution regional models. ERA5 is the international standard for historical atmospheric reanalysis, covering the globe at approximately 31-km horizontal resolution from 1940 to present. Open-Meteo re-processes this reanalysis and delivers it through a simple REST API without authentication or cost.

The archive endpoint accepts a latitude/longitude pair, a date range, and a list of variables, and returns a JSON payload of daily aggregates. One API call was made per airport coordinate. The responses were cached to disk so that re-running the pipeline incurs no additional requests.

2.2.2 Why Open-Meteo over alternatives

Three considerations drove the choice of Open-Meteo over PRISM, the leading U.S. gridded climatology product:

1. **Variable coverage.** Open-Meteo returns temperature, precipitation, snowfall, wind speed, and humidity in a single call. PRISM provides only temperature and precipitation, omitting the humidity and wind channels relevant for apparent-temperature analysis.
2. **Spatial match to EV Watts.** EV Watts identifies sessions only at the metro-area level, not at individual station coordinates. PRISM's 4-km grid therefore offers no practical advantage over a point query at the metro's representative coordinate.
3. **Engineering cost.** PRISM data are distributed as gridded raster files (.bil) requiring rasterio or GDAL. Open-Meteo requires one HTTP request per location, reducing data-acquisition code to under 50 lines.

2.2.3 Spatial assignment: airport coordinates

Because EV Watts does not provide charger-level coordinates, one representative weather location per metro must be chosen. We assign the primary commercial airport of each metro rather than a city-center coordinate, for three reasons.

First, U.S. NWS official metropolitan climate norms are reported at ASOS stations co-located at major airports (ORD for Chicago, LAX for Los Angeles, LGA for New York). Using airport coordinates aligns our weather series with the same reference stations used in official climatological statistics.

Second, airports are sited away from dense urban cores, reducing urban heat-island contamination; the recorded temperature therefore better approximates ambient conditions across the metro commute-shed where most EV charging occurs.

Third, the metro fixed effects used in the analysis absorb the permanent level difference between the assigned airport temperature and the true metro-average temperature. What matters is that the airport time series tracks the metro average, which it does because synoptic weather systems move across entire metros simultaneously.

2.2.3.1 Multi-point averaging for microclimate metros.

Two metros exhibit intra-metro climate gradients large enough to make a single airport unrepresentative. For Los Angeles–Long Beach–Anaheim, CA, the coastal marine layer produces systematically cooler conditions near the coast (LAX) compared with the inland San Gabriel Valley (BUR) and South Bay (LGB). All three airports are queried and their daily values averaged. For Seattle–Tacoma–Bellevue, WA, a mild gradient between Sea-Tac (SEA) and Boeing Field (BFI) motivated a two-point average. All other metros are assigned a single airport.

Table 5 lists all 29 metros, the airport(s) assigned, and the coordinates queried.

Table 5: Metro areas, assigned airports, and query coordinates.

Metro Area (EV Watts label)	Airport	Lat	Lon
Albany-Schenectady-Troy, NY	ALB	42.748	−73.802
Ann Arbor, MI	ARB	42.223	−83.745
Austin-Round Rock-Georgetown, TX	AUS	30.197	−97.666
Baltimore-Columbia-Towson, MD	BWI	39.177	−76.668

Boston-Cambridge-Newton, MA-NH	BOS	42.366	−71.020
Boulder, CO	BDU	40.039	−105.226
Burlington-South Burlington, VT	BTV	44.472	−73.153
Chicago-Naperville-Elgin, IL-IN-WI	ORD	41.978	−87.904
Dallas-Fort Worth-Arlington, TX	DFW	32.897	−97.038
Denver-Aurora-Lakewood, CO	DEN	39.857	−104.673
Des Moines-West Des Moines, IA	DSM	41.534	−93.660
Detroit-Warren-Dearborn, MI	DTW	42.213	−83.353
Grand Rapids-Kentwood, MI	GRR	42.880	−85.522
Kansas City, MO-KS	MCI	39.297	−94.714
Las Vegas-Henderson-Paradise, NV	LAS	36.080	−115.152
Los Angeles-Long Beach-Anaheim, CA	LAX / BUR / LGB	33.943 / 34.201 / 33.818	−118.408 / −118.359 / −118.152
Miami-Fort Lauderdale-Pompano Beach, FL	MIA	25.793	−80.291
New York-Newark-Jersey City, NY-NJ-PA	LGA	40.777	−73.872
Philadelphia-Camden-Wilmington, PA-NJ	PHL	39.872	−75.241
Phoenix-Mesa-Chandler, AZ	PHX	33.434	−112.012
Pittsburgh, PA	PIT	40.491	−80.233
Portland-Vancouver-Hillsboro, OR-WA	PDX	45.589	−122.595
Providence-Warwick, RI-MA	PVD	41.724	−71.428
Reno, NV	RNO	39.499	−119.768
Rochester, NY	ROC	43.118	−77.672
Salem, OR	SLE	44.909	−123.002
Seattle-Tacoma-Bellevue, WA	SEA / BFI	47.450 / 47.530	−122.309 / −122.302
Washington-Arlington-Alexandria, DC	DCA	38.852	−77.038
Worcester, MA-CT	ORH	42.267	−71.875

2.2.4 Variables retrieved

Ten daily weather variables were requested for each coordinate over 2019-06-25 to 2022-12-31. Table 6 describes each variable and its unit.

Table 6: Daily weather variables retrieved from Open-Meteo.

Variable	Unit	Description
temperature_2m_max	°C	Daily maximum air temperature at 2 m
temperature_2m_min	°C	Daily minimum air temperature at 2 m
temperature_2m_mean	°C	Daily mean air temperature at 2 m
precipitation_sum	mm	Total daily precipitation (rain + snow water equivalent)
snowfall_sum	cm	Daily snowfall depth

2.2.5 Acquisition pipeline and caching

API responses were cached as individual JSON files (e.g., ORD.json, LAX.json). Before each request the pipeline checks whether a cache file exists; if so, it reads from disk. A 20-second sleep was inserted between requests to respect the free service tier's rate limits. For Los Angeles and Seattle, each airport was queried separately; the resulting daily DataFrames were concatenated and the unweighted mean of all numeric variables was computed by date, producing one metro-averaged series. After all metro series were assembled, they were concatenated into a single tidy DataFrame keyed on metro_area and date, sorted, and written to data/processed/weather_daily.parquet. Table 7 summarises the output dataset.

Table 7: Properties of weather_daily.parquet.

Property	**Value**
Rows (metro × date)	29 × 1,286 = 37,294
Columns	12 (metro_area, date, 10 weather vars)
Date range	2019-06-25 to 2022-12-31
Number of metros	29
File format	Apache Parquet
Missing values	None (ERA5 reanalysis is gap-free)

2.3 Merging EV and Weather Data

2.3.1 Input files

The merge step combines the two upstream datasets:

- EV merged.parquet — the cleaned session-level EV Watts file (one row per charging session), with metro area and date derived from start datetime.
- weather daily.parquet — the metro-day weather panel (Section 2), one row per (metro area, calendar date) pair with ten daily weather variables.

Both files had their date columns cast to datetime64 prior to joining to ensure a type-safe merge.

2.3.2 Join design

The two datasets were joined using a left join on the composite key (metro_area, date). The left join preserves every row in the EV session file; sessions that do not match a weather record receive NaN for all weather columns. A Pandas indicator column was used to verify the join quality. The hit rate was computed as:

$$hit\ rate = \frac{row\ with\ \_merge{=}both}{total\ rows} \times 100\%$$

A hit rate near 100% is expected because the weather panel covers all 29 named metros for every day in the EV Watts date range; deviations flag sessions from unmapped metros or dates outside 2019-06-25 to 2022-12-31.

2.3.3 Output files

The merge step produces two analytical files. The first is a session-level analytical dataset. The merged DataFrame was saved as data/processed/merged_metro_day.parquet. Despite the file name, this is a session-level dataset — one row per charging session with ten weather columns appended. It is the primary input for all downstream figures and models. Table 8 summarizes the file.

Table 8: Properties of merged_metro_day.parquet.

Property	Value
Granularity	One row per charging session
Left key	EV Watts cleaned sessions (all retained)
Right key	weather_daily.parquet
Join columns	metro_area, date
Columns added	10 weather variables

The second is an aggregated metro–day–charger-level panel, which was constructed by grouping the merged session file by (metro_area, date, charge_level). Table 9 lists the aggregation operations applied.

Table 9: Aggregation operations for metro_day_charger_panel.parquet.

Source column	Aggregation	Output column
session_id	count	sessions
energy_kwh	sum	energy_kwh_sum
energy_kwh	mean	energy_kwh_mean
charge_duration	mean	charge_duration_mean
10 weather variables	first	retained as-is

Weather variables are constant within a (metro, date) group, all sessions on the same metro-day share the same weather record, so first is the correct aggregator and introduces no information loss. The panel was saved as data/processed/metro_day_charger_panel.parquet.

3. Infrastructure Overview

3.1 EV Charging Network Density and Utilization

To investigate the EV charging network density and utilization, we took the session-level data and grouped it by metro_area. Then, we calculated rates to measure performance. The final three metrics were:

- Total Number of EVSEs: The count of charging ports in each matro_area
- Utilization (Sessions / EVSE): The average number of charging sessions handled by each port.
- Energy Intensity (Energy / EVSE): The average amount of Total energy delivered during the session per port.
- Avg Energe /session: total energy divided by total number of sessions per metra area.

As we cleaned the metro area names to match a reference dataset, allowing us to merge in the precise Latitude and Longitude coordinates for each city.

Figure 1 presents the geographic distribution of EV charging infrastructure and utilization rates across 29 U.S. metropolitan areas over the 2020–2022 period. Figure 1 presents two panels: total EVSE count per metro (Panel 1A) and utilization intensity measured as charging sessions per EVSE (Panel 1B).

In terms of infrastructure density, Portland-Salem emerges as the most heavily equipped metro area in the dataset, followed by Washington D.C. and Detroit as secondary hubs. This concentration of infrastructure in the Pacific Northwest and parts of the Midwest and Mid-Atlantic reflects early public investment in EV charging networks in these regions. By contrast, interior metros including Kansas City, Des Moines, and Reno exhibit notably sparse EVSE deployment, suggesting uneven diffusion of charging infrastructure across the continental U.S.

Panel 1B reveals a markedly different geographic pattern when infrastructure is weighted by usage intensity. Los Angeles records the highest utilization rate, exceeding 900 sessions per EVSE, indicating that existing chargers are under considerable demand pressure relative to network size. Phoenix and Austin similarly display disproportionately high utilization despite comparatively smaller installed bases, consistent with a scenario where infrastructure expansion has not kept pace with growing EV adoption in Sun Belt markets. Conversely, Washington D.C. and Detroit exhibit only moderate utilization rates, despite ranking among the top metros by total EVSE count, suggesting that supply in these areas is broadly adequate relative to current demand levels.

Taken together, Figure 1 highlights a structural divergence between infrastructure-rich, moderately utilized metros and infrastructure-constrained, heavily utilized ones, showing a distinction that has direct implications for how weather-driven variation in charging demand manifests across different market contexts.

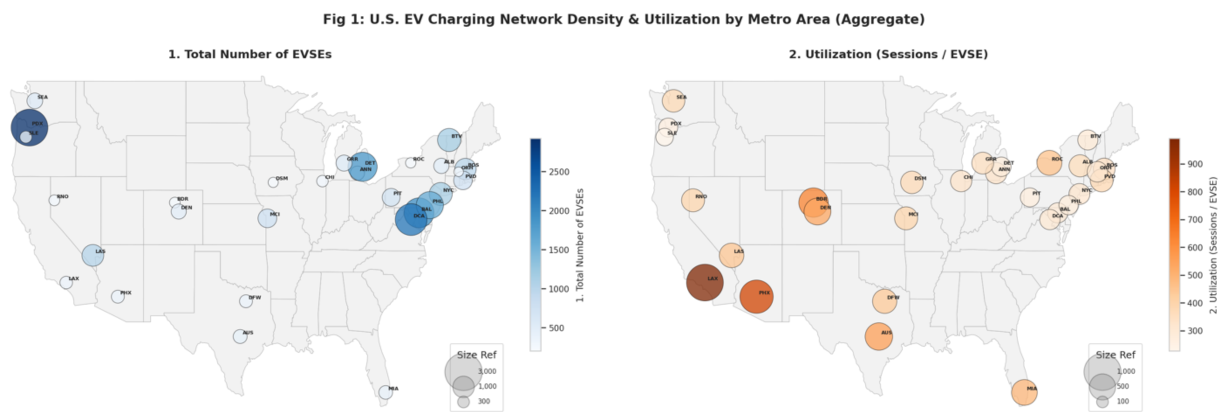

Figure 1: U.S. EV Charging Network Density and Utilization by Metro Area (Aggregate)
Panel 1A: Total number of EVSEs by U.S. metropolitan area, 2020–2022.
Panel 1B: Utilization intensity (charging sessions per EVSE) by U.S. metropolitan area.

3.2 Energy Intensity and Average Session Depth

Figure 2 extends the analysis by examining two complementary dimensions of energy consumption: total energy throughput per charger (Panel 2A) and average energy delivered per individual charging session (Panel 2B). Together, these panels provide insight into whether observed differences in charging demand are driven by session frequency, session depth, or both.

Panel 2A maps energy intensity, measured as total kilowatt-hours delivered per EVSE over the study period. Los Angeles and Phoenix record the highest energy intensity values, exceeding 22,500 kWh per EVSE, while Dallas-Fort Worth and Austin also rank prominently. This Sun Belt concentration suggests that thermal stress on vehicle batteries, leading to greater cumulative energy throughput per charger, which may be driven by sustained high ambient temperatures.

Panel 2B shifts focus to average energy per session, revealing a distinct geographic pattern. Miami records the highest per-session energy draw, approaching 30 kWh per session, with several Northeast corridor metros also clustering at the upper end of the distribution. Cold-weather markets in the Northeast likely experience elevated per-session energy demand due to well-documented reductions in lithium-ion battery efficiency at low temperatures. Concurrently, warm and hot southern metros such as Miami, Phoenix, and Dallas-Fort Worth also exhibit above-average per-session values, consistent with increased auxiliary energy loads under high ambient temperatures through using air condition.

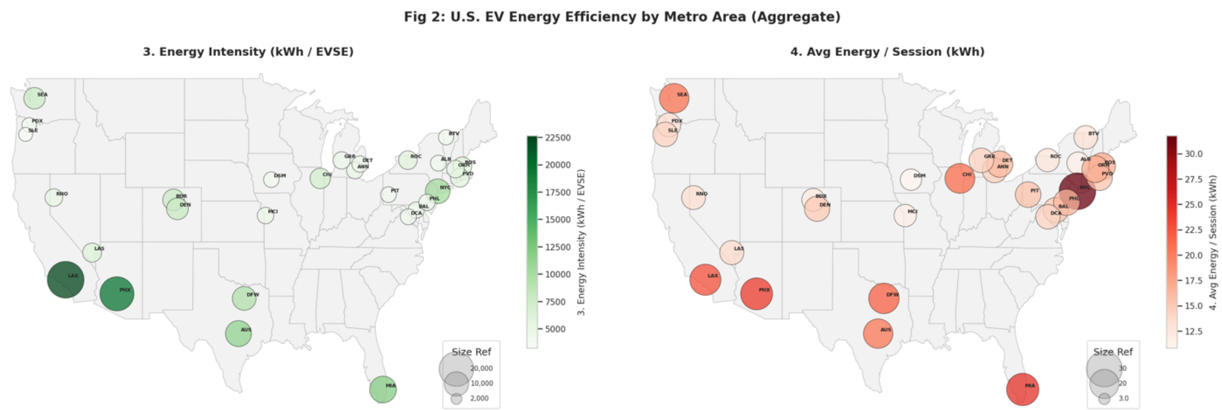

Figure 2: U.S. EV Energy Efficiency by Metro Area (Aggregate)
Panel 2A: Energy using efficiency by metro area
Panel 2B: The average energy using efficiency per session

Collectively, Figures 1 and 2 reveal that both thermal extremes — cold and heat — amplify EV energy demand, albeit through distinct mechanisms. This geographic heterogeneity provides empirical motivation for examining daily weather variation as a systematic driver of charging behavior in the subsequent analysis.

4. Weather and charging behavior: descriptive evidence

We examine two complementary channels through which weather can shape EV charging activity. The intensive margin asks how much energy flows per session, conditional on a session occurring; the extensive margin asks how many sessions occur in the first place. Figure 3 isolates the first channel by relating energy per session to ambient temperature; Figures 4 and Figure 5 isolate the second channel by relating session counts to precipitation and snowfall, respectively. The contrast across the three figures is the central finding of this study: the physical channel is always on and largest under cold temperatures, while the behavioral channel is muted in rain but very large in heavy snow.

4.1 Intensive Margin: Temperature and kWh per Session

4.1.1 Approach

For each U.S. metropolitan area and calendar date, we compute the mean energy delivered per charging session by averaging energy_kwh across all sessions that started in that metro on that day. We discard metro-days with fewer than three sessions or missing weather. The resulting panel contains 34,424 metro-day observations across 29 metros from 2019-06-25 through 2022-12-31.

We then isolate the within-metro effect of weather by residualizing the kWh-per-session series against a metro× month-of-year fixed effect, adding the grand mean back so the y-axis remains in interpretable kWh units, as opposed to permanent differences between metros(Chicago drivers charge differently from Phoenix drivers) and seasonal patterns (everyone charges differently in December than in July).

Metro-days are then binned into 2 °C intervals of daily mean temperature. We plot the bin means with 95% confidence intervals and overlay a weighted quadratic fit (weights = bin counts) to characterize the shape of the relationship and locate the vertex.

4.1.1 Result

Figure 3 reveals a clear and asymmetric U-shaped relationship between daily mean temperature and energy delivered per session. The fitted quadratic minimum sits at 14.1 °C, at the upper edge of what battery engineers describe as the lithium-ion thermal comfort zone.

From Figure 3, the cold-side response is sharp. As the daily mean temperature falls from 0 °C to −16 °C, mean energy per session rises from roughly 16.3 kWh to 18.2 kWh, above the comfort-zone baseline of ~15.9 kWh. By contrast, the hot-side response is more modest. From 25 °C upward, energy per session creeps up by roughly 6%, peaking near 34 °C at about 16.8 kWh.

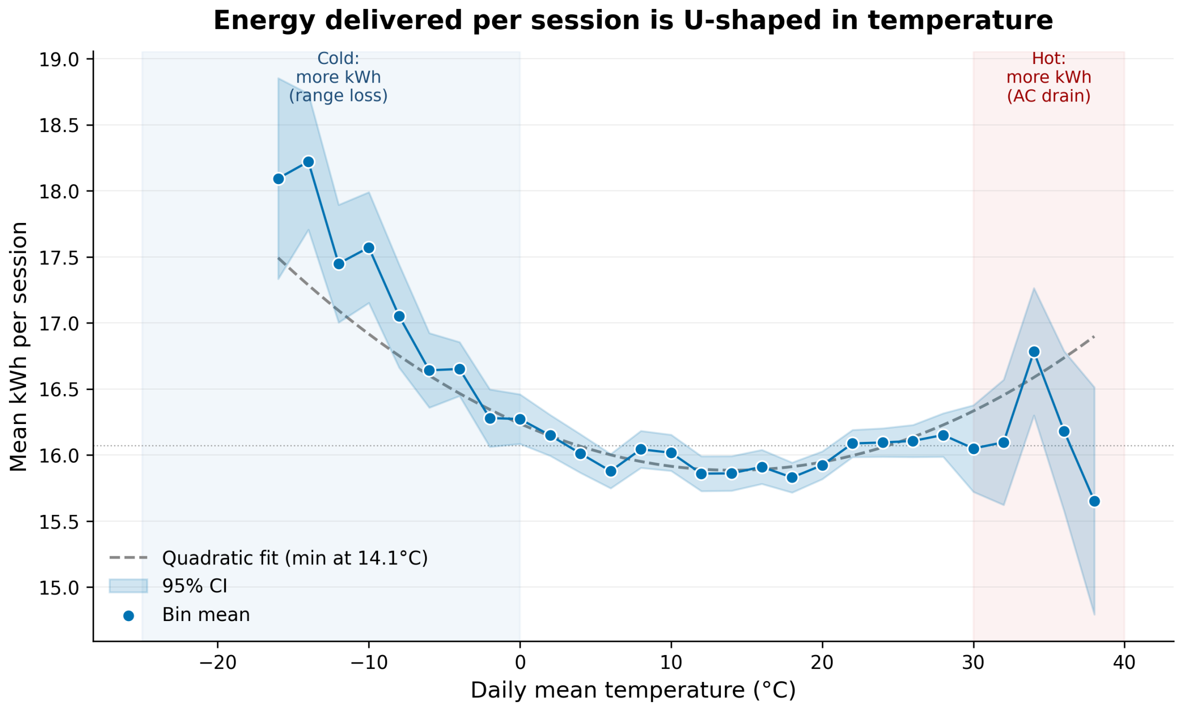

Figure 3: Mean kWh delivered per session vs. daily mean temperature, demeaned by metro × month-of-year. Bands show 95% confidence intervals; dashed line is a weighted quadratic fit. Sample: 34,424 metro-days across 29 metros, 2019-06-25 to 2022-12-31.

4.1.3 Interpretation

4.1.3.1 Cold-weather battery range loss

On the cold side, lithium-ion battery capacity and discharge efficiency fall sharply below roughly 5 °C as electrolyte conductivity drops, internal resistance rises, and cabin heating draws several kilowatts directly from the traction battery. Real-world studies (AAA, Recurrent, Geotab) consistently document EV range losses of 20–40% at sub-freezing temperatures, which translates to a 10–20% increase in kWh per session if drivers maintain the same daily mileage. Our 14% estimate at −16 °C falls cleanly within this engineering range.

4.1.3.2 Hot-weather AC load.

On the hot side, cabin air conditioning draws 1–3 kW continuously above 30 °C, and the battery itself is somewhat less efficient at very high temperatures; together these produce a smaller but real range penalty consistent with our 6% estimate.

4.1.3.3 Interpretation of finding

The gap between the two arms is a feature, not a defect. Cold temperatures reduce capacity through mechanisms such as slower ion transport, pre-heating losses, and increased internal resistance. In contrast, heat in the typical U.S. ambient range mainly contributes by adding an AC load. Therefore, the observed U-shape, which is not symmetric, aligns with the engineering literature. The minimum point at 14 °C is slightly below typical cabin comfort (~21 °C). This makes sense because the determining factor for kWh per session is the battery, not the driver. Thus, the vertex at 14 °C reflects the lithium-ion thermal optimum.

4.2 Extensive Margin: Precipitation and Session Count

4.2.1 Approach

Where Figure 3 measured the energy of each session, Figure 4 measures how many sessions occur. Using the same metro-day panel, we compute the daily session count for each metro-day. To remove permanent differences across metros and seasonal cycling, we express each metro-day's count as a ratio of the (metro × month-of-year) mean. A ratio of 1.0 corresponds to a typical day for that metro in that month. Values below 1.0 indicate fewer sessions than normal; values above 1.0 indicate more.

Daily precipitation totals from Open-Meteo are binned into five NOAA-aligned categories: None (= 0 mm), Light (≤ 2.5 mm), Moderate (2.5–10 mm), Heavy (10–25 mm), and Extreme (> 25 mm). We display the within-category ratio distribution as a notched boxplot (Figure 5).

4.2.2 Result

The data show a small but coherent extensive-margin response to precipitation. Moving from dry days to extreme-rain days, the median ratio of sessions to the metro-month mean falls from 0.91 to 0.87, a reduction of roughly 4%. The notch on the Extreme category (n = 710) sits clearly below the notch on None (n = 17,687), indicating that the difference is statistically distinguishable. The Light category, by contrast, is indistinguishable from None (median 0.91 in both, +0.0%): days with under 2.5 mm of precipitation look behaviorally identical to dry days.

The overall pattern suggests a threshold structure rather than a smooth dose-response. Dry and trace days drive nearly identical volumes; once precipitation crosses ~2.5 mm, median volume drops by 3–4% and stays at that lower level through the heavy and extreme bins.

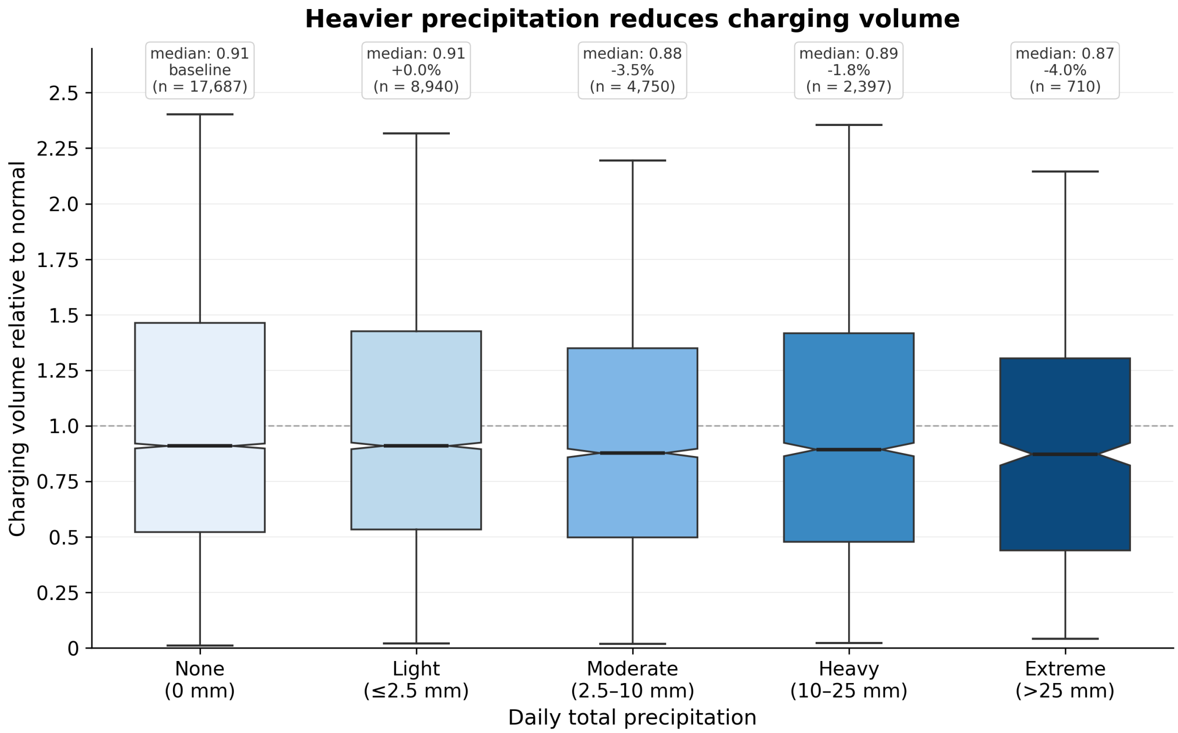

Figure 4: Daily session count expressed as a ratio of the metro × month-of-year mean, by precipitation category (NOAA 5-bin). Notched boxplots; whiskers shortened to 1.0×IQR for visual clarity. Sample: 38,484 metro-days across 29 metros.

4.2.3 Interpretation

The small magnitude of the precipitation response is consistent with the underlying economics of EV charging. Most U.S. public and workplace charging is necessity-driven: people charge so they can commute tomorrow, not because they are taking discretionary trips that can be canceled when the weather turns bad. Heavy rain may suppress some discretionary errands and shopping runs, but it does not stop people from going to work or returning home, and these routine trips dominate our sample. The 4% reduction we observe on extreme-rain days is therefore a plausible upper bound on the share of charging activity that is genuinely rain-elastic.

The flat plateau across the Moderate, Heavy, and Extreme bins is itself informative. The response looks largely on/off rather than dose-response, consistent with a threshold model in which drivers either decide that rain is bad enough to defer a trip or press through, with the precise intensity mattering less than whether the day is "rainy" at all.

4.3 Extensive Margin: Snowfall and Session Count

4.3.1 Approach

Figure 5 applies the same framework as Figure 4 to snowfall instead of rain. Two design choices differ. First, because snow is concentrated in northern metros, we restrict the analysis to the 23 "snow-prone" metros that experience at least 30 snow days in the panel; this excludes desert and Sun Belt metros (Phoenix, Miami, Los Angeles, etc.) where snow events are atypical and would distort the dry-day baseline. Second, because heavy-snow days are scarcer than heavy-rain days, we use four NOAA-aligned categories rather than five: None (= 0 cm), Light (≤ 2.5 cm), Moderate (2.5–10 cm), and Heavy (> 10 cm). The visualization style (notched boxplot, identical y-axis range and ticks) matches Figure 4 to make the rain-vs-snow contrast directly readable in side-by-side viewing.

4.3.2 Result

The snowfall response is an order of magnitude larger than the rainfall response, and the shape is qualitatively different. Light snow (≤ 2.5 cm, n = 2,442) is statistically indistinguishable from dry days (median 0.90 vs. 0.91, a difference of −1.2%). Once snowfall crosses 2.5 cm, however, the response escalates rapidly. Moderate snow days (2.5–10 cm, n = 555) show a median of 0.76, a reduction of −16.6%; Heavy snow days (> 10 cm, n = 84) show a median of just 0.44, a reduction of −51.4% relative to the dry-day baseline. The pattern is monotonically increasing in event severity, forming a clean dose–response.

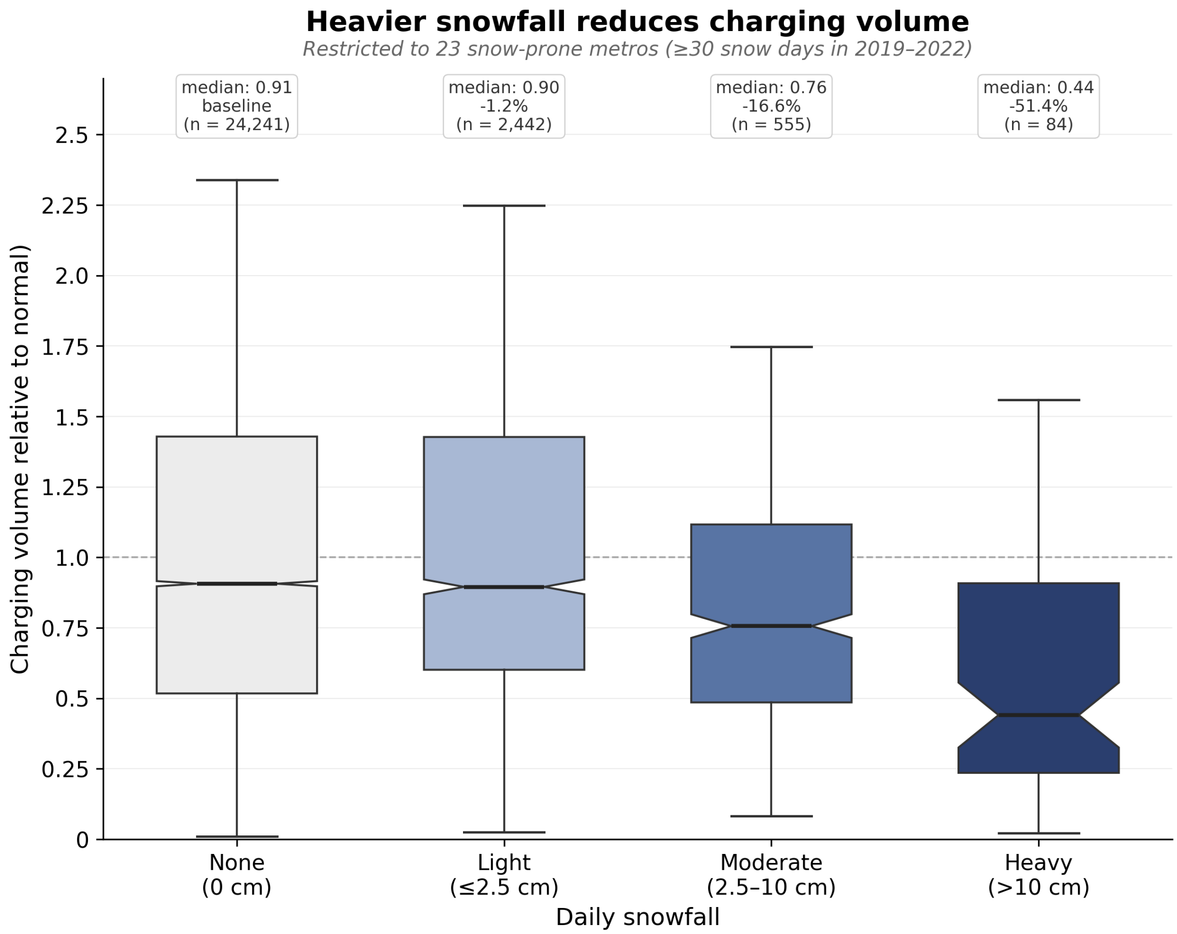

Figure 5: Daily session counts relative to the metro × month-of-year mean, by snowfall category. Sample restricted to 23 snow-prone metros (≥30 snow days in 2019–2022). Y-axis matches Figure F4 for direct visual comparison.

4.3.3 Interpretation

The contrast with Figure 4 reframes Figure 5 findings. Charging volume is not generally weather-inelastic; it is rain-inelastic. When the weather event is severe enough to genuinely impair driving as snow does, the behavioral response becomes dramatic.

Two mechanisms operate simultaneously once snowfall crosses the moderate threshold. The first is a road-safety mechanism: driving in snow is dangerous, so discretionary trips are deferred and some commutes are canceled outright. The second is an institutional-cancellation mechanism: employers, schools, and municipalities trigger closures above certain thresholds, mechanically reducing the number of trips that can occur. Both reinforce each other and switch on together, producing the steep monotone decline visible in the figure.

The result also rationalizes the apparent inelasticity in Figure 4: rain produces inconvenience but not disruption, and U.S. EV drivers, like U.S. gasoline drivers, push through inconvenience because their trips are mostly necessities. Heavy snow cuts charging volume roughly in half, which leads to an effect more than twelve times larger than the corresponding effect for extreme rain.

4.4 Synthesis

Figures 3, 4, and 5 together report weather sensitivity in EV charging through two channels of very different character. The physical channel (Figure 3) operates through battery chemistry and HVAC load, is always on, and produces a roughly 14% increase in energy per session under extreme cold. The behavioral channel (Figures 4 and 5) operates through driver decisions and institutional closures, but switches on only when the weather event is severe enough to materially disrupt driving: extreme rain reduces daily session counts by just 4%, while heavy snow cuts them by more than 50%. The first channel reflects how EVs work; the second reflects how, and whether, drivers can use them. The disparity in magnitudes tells us that EV charging in this period is far more sensitive to the temperature of the air than to the form or quantity of precipitation, with one striking exception when the precipitation falls as snow.

5. Weather and Charging behavior: econometric evidence

This section examines the relationship between weather conditions and EV charging behavior using additional descriptive figures and pooled OLS specifications. Figures 6 through 8 characterize charger-type heterogeneity, calendar structure, and within-metro weather correlations. The T2.1 regressions then summarize selected relationships in a fixed-effects framework. Across these analyses, weather is associated with charging behavior, but the relationship appears modest, nonlinear, and strongly conditioned by charger type and calendar timing.

5.1 L2 vs. DCFC Weather Heterogeneity

Figure 6 examines whether the weather-charging relationship differs between L2 and DCFC. This distinction is substantively important because the two charger types are associated with different charging contexts. L2 charging is more likely to occur during routine or long-dwell parking, including workplace, residential, and daytime parking. DCFC is more likely to reflect opportunistic or trip-related charging. If weather affects travel demand, battery range, or the cost of stopping outdoors, the charging response may differ across charger types.

5.1.1 Approach

The analysis uses the metro-day-charger panel, where each observation is a metro-area by date by charge-level cell. Daily mean temperature is grouped into 2 °C bins. For each bin and charger type, the plotted point is the average demeaned charging outcome for both mean energy per session and session count, adjusted for metro and month-of-year patterns within charger type.

5.1.2 Result

The mean kWh panel shows a mild U-shaped pattern: energy delivered per session is higher on colder and hotter days and lower around moderate temperatures. This is consistent with the idea that vehicle energy needs may be higher under thermal stress. The session-count panel is less smooth, especially for DCFC, suggesting that temperature may affect energy per session more clearly than the number of plug-in events. Overall, Figure 6 supports the idea that weather responsiveness differs by charger type, but the evidence remains descriptive rather than causal.

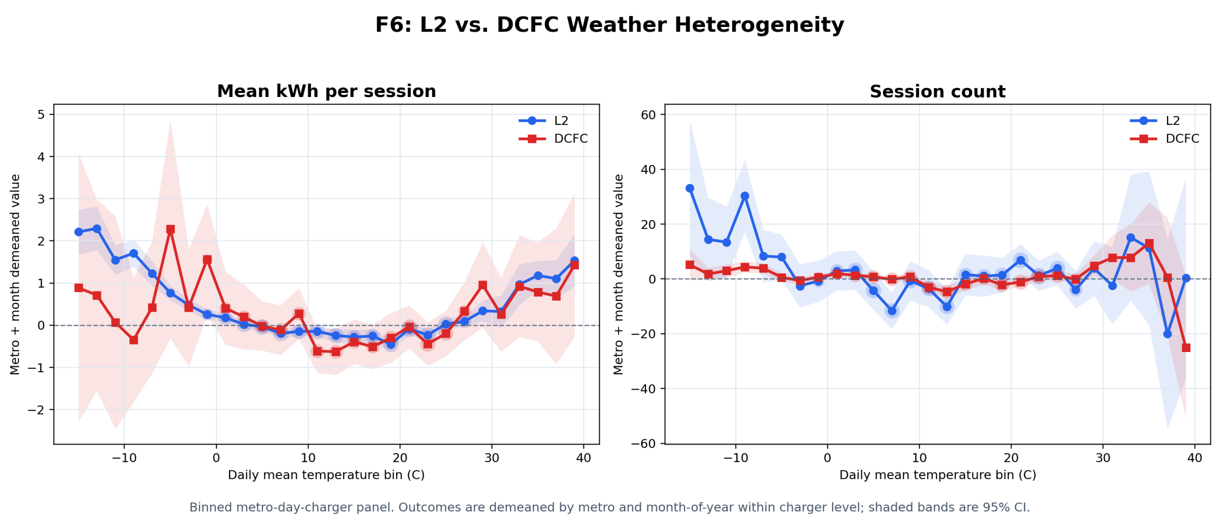

Figure 6: Binned relationship between daily mean temperature and charging outcomes, separately for Level 2 and DC fast charging.

5.2 Calendar Heatmap of Charging Demand

5.2.1 Approach

Figure 7 documents the calendar structure of charging demand. This timing pattern is relevant because estimated weather relationships are observed on top of systematic variation by day of week and hour of day. Strong calendar regularities imply that weather-related variation should be interpreted relative to underlying routines in driving, parking, and charging behavior.

The analysis uses session-level charging records with start timestamps. For each session, the day of week and hour of day are extracted, and sessions are counted by charger type, date, day of week, and hour. These counts are averaged across dates to create a 7 × 24 calendar heatmap for each charger type. Darker cells indicate more charging sessions.

5.2.2 Result

The L2 panel shows a clear weekday daytime pattern, which is consistent with routine charging linked to work, home, or other long parking periods. DCFC has lower total volume and a more concentrated midday and afternoon pattern, which is more consistent with opportunistic or trip-related charging. This matters for the broader analysis because weather effects do not operate on a blank baseline; they interact with when people are likely to drive, park, and charge.

The main limitation is that the heatmap averages over all metros and dates. It therefore compresses local variation and does not control growth in the charging network over time. The heatmap should therefore be interpreted as descriptive evidence on the calendar rhythm of charging demand rather than as an estimate of weather responsiveness.

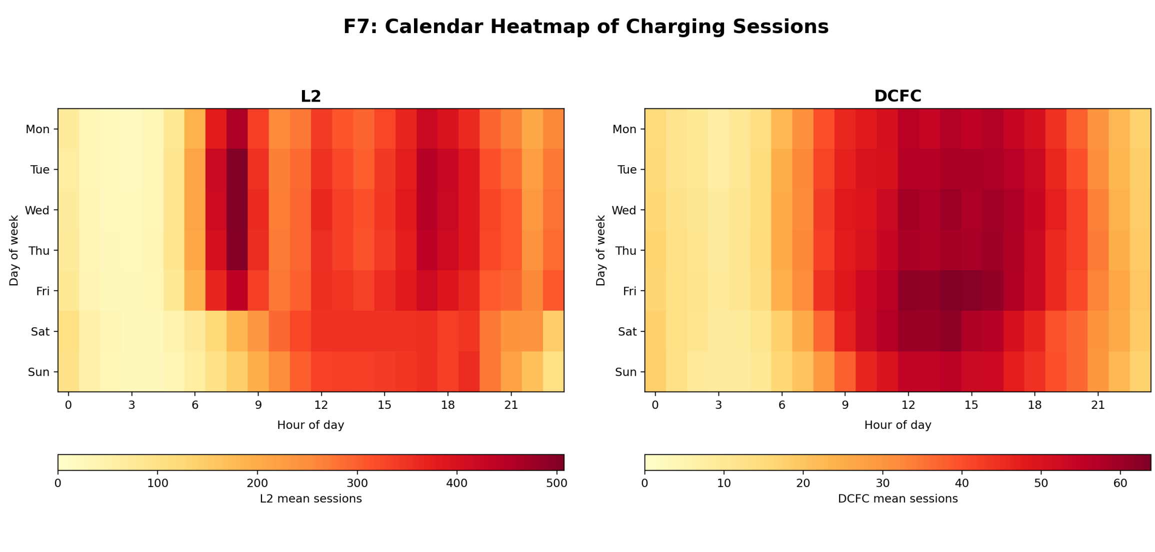

Figure 7: Calendar heatmap of charging sessions by day of week and hour of day, separately for L2 and DCFC.

5.3 Metro-Demeaned Weather-Charging Correlation Matrix

Figure 8 reports Pearson correlations between weather variables and charging outcomes, computed separately for L2 and DCFC after removing metro-level means. This provides a descriptive summary of the direction and magnitude of within-metro weather-charging associations.

5.3.1 Approach

The weather variables include daily mean temperature, maximum temperature, minimum temperature, precipitation, humidity, and wind speed. The charging outcomes include session count, total kWh, mean kWh per session, and mean charging duration. Before computing the correlations, each variable is demeaned within metro and charger type, so the correlations are based on day-to-day variation within the same metro rather than differences between colder and warmer cities.

5.3.2 Result

The correlations are generally small in magnitude. This suggests that weather is not the dominant driver of daily charging outcomes in the panel. Calendar patterns, adoption growth, charger availability, and local charging routines are likely to explain much more of the variation. However, small average correlations do not rule out nonlinear relationships, which motivates the binned temperature analysis and the quadratic temperature specification.

The main limitation is that Pearson correlations are linear summaries. They can miss U-shaped or threshold relationships, especially for temperature. Figure 8 should therefore be interpreted as a diagnostic summary of direction and magnitude, not as a causal model.

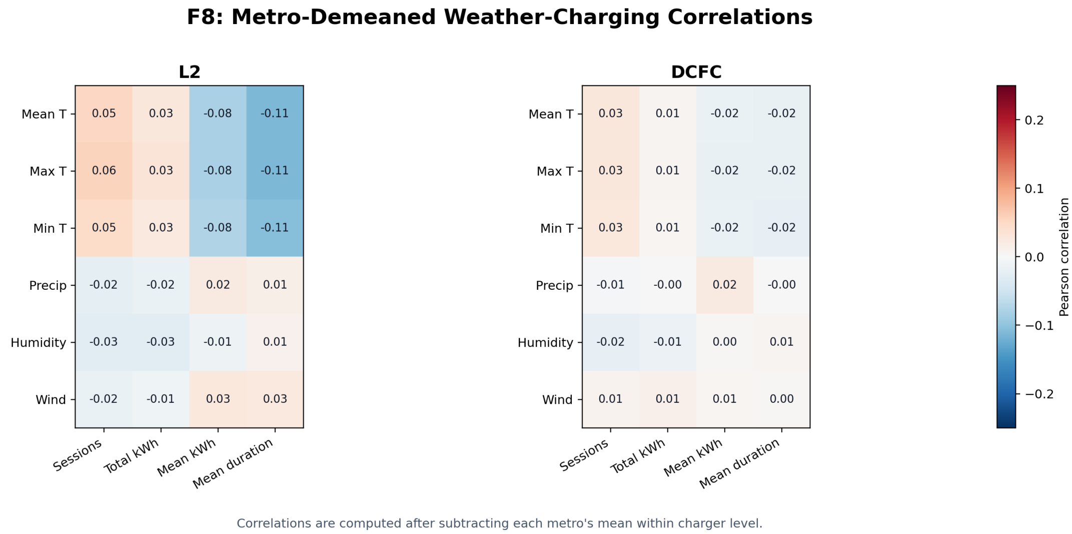

Figure 8: Pearson correlations between weather variables and charging outcomes after demeaning within metro and charger type.

5.4 Econometric Specification

The econometric analysis estimates pooled OLS models on a metro-day panel constructed from the charging data. The specifications are descriptive and are intended to quantify selected associations between weather variables and charging outcomes after accounting for metro and calendar fixed effects. Dependent variables are log-transformed, and standard errors are heteroskedasticity-robust.

5.4.1 Model specifications

Model 1: energy per session

$$\log(\overline{kWh}_{mt}) = \beta_1\, Temp_{mt} + \beta_2\, Temp^2_{mt} + \alpha_m + \gamma_{ym} + \varepsilon_{mt}$$

Model 2: precipitation and session count

$$\log(Sessions_{mt}) = \Sigma b\ \delta b \cdot 1\{PrecipBin_{mt}=b\} + \alpha_m + \lambda dow + \varepsilon_{mt}$$

Model 3: combined session-count specification

$$\log(Sessions_{mt}) = \beta_1\, Temp_{mt} + \beta_2\, Temp^2_{mt} + \Sigma b\ \delta b \cdot 1\{PrecipBin_{mt}=b\} + \alpha_m + \gamma_{ym} + \lambda dow + \varepsilon_{mt}$$

5.4.2 Regression results

Table 10: Pooled OLS models with robust HC1 standard errors in parentheses.

Variable	(1) log(mean kWh)	(2) log(sessions)	(3) log(sessions)
Mean temperature (°C)	−0.0079*** (0.0002)		0.0026*** (0.0006)
Mean temperature squared	0.0002*** (0.0000)		−0.0000** (0.0000)
Light precipitation (<2mm)		−0.0236** (0.0094)	−0.0144*** (0.0043)
Moderate precipitation (2–10mm)		−0.0555*** (0.0108)	−0.0374*** (0.0049)
Heavy precipitation (>10mm)		−0.0689*** (0.0140)	−0.1062*** (0.0071)
Metro fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	No	Yes
Day-of-week fixed effects	No	Yes	Yes
Implied U-shape minimum (°C)	20.02		
Observations	34,523	34,523	34,523
R-squared	0.810	0.505	0.896

*Significance: *** p<0.01, ** p<0.05, * p<0.10.*

Model 1 regresses log mean kWh per session on daily mean temperature and its square, with metro and year-month fixed effects. The implied minimum of the fitted quadratic is $-\beta_1 / (2\beta_2) = 20.02$ °C.

Model 2 regresses log session count on precipitation bins, with metro and day-of-week fixed effects. The omitted category is no precipitation. The bins separate light precipitation below 2 mm, moderate precipitation between 2 and 10 mm, and heavy precipitation above 10 mm.

Model 3 is the combined specification for log session count, including the temperature quadratic and precipitation bins together with metro, year-month, and day-of-week fixed effects. This specification evaluates whether precipitation-bin estimates remain negative after controlling for temperature and monthly seasonality.

The estimates indicate that the temperature quadratic for log mean kWh implies a U-shaped relationship with the predicted minimum near 20.02 °C. For session counts, precipitation-bin coefficients are negative, and the negative association is largest for heavy precipitation in the combined model. These estimates should be interpreted as descriptive rather than causal because the specifications do not isolate random weather shocks. They nevertheless provide fixed-effects evidence consistent with the descriptive patterns shown in Figures 6 through 8.

6. Conclusion

This study examines how ambient weather shapes EV charging demand across 29 U.S. metropolitan areas from 2020 to 2022, finding evidence of two distinct channels that operate at different magnitudes and under different conditions affecting the EV charging behavior.

The first channel is continuously active and operates through battery chemistry and the load of air conditioning. Energy delivered per session follows an asymmetric U-shaped relationship with temperature, with the minimum occurring around 14–20 °C. Cold temperatures substantially increase charging demand due to the decrease of battery efficiency, while hot temperatures have smaller effects which are mainly affected by additional air-conditioning demand.

The second channel operates mainly through driver behavior and individual responses and becomes important only during severe weather events. Rainfall has only modest effects on charging activity, but snowfall causes major disruptions, the moderate snow significantly reduces session counts, while heavy snow cuts charging activity by more than half. This contrast suggests that snow affects charging through both road-safety concerns and institutional closures.

Additional analyses show that weather responses also differ between L2 and DCFC, reflecting their distinct use cases and individual choices. The regression results could be related to descriptive evidence like the U-shaped temperature relationship and suggests the increasingly negative effects of severe precipitation after controlling for metro and calendar fixed effects.

These findings have practical implications for charging infrastructure planning. In cold temperature weather conditions, the main challenge is increased quality of battery rather than insufficient charging ports, since cold weather raises energy demand without substantially reducing session frequency. By contrast, heavy snow days have great limitations on charging behavior because charging activity declines sharply.

This study also has several limitations. The data cover only 29 metropolitan areas, and the research period covers only 2019–2022, which may limit broader generalizability. Weather conditions are measured at the metro level using airport coordinates, potentially introducing measurement errors. And this study only investigates the relationship between weather conditions and the charging behavior, ignoring other latent factors, such as the demand of charging also increased by the holidays. In addition, the OLS estimates are descriptive rather than causal and may reflect unobserved seasonal confounders. Future research could strengthen causal identification strategies, incorporate behavioral and socioeconomic controls, and examine a broader geographic and temporal scope to improve validity.

Overall, the paper provides systematic empirical evidence on how weather shapes both the frequency and intensity of EV charging demand, offering insights relevant to grid planning, charging infrastructure design, and future EV adoption.

Contribution statement: Jiahan Yin was responsible for cleaning and preprocessing the EV charging data and producing the aggregate infrastructure visualizations (Figures 1 and 2). Chen Qing was responsible for cleaning the weather data, merging the EV charging and weather datasets, and constructing the weather-response visualizations (Figures 3, 4, and 5). Jinyang Li was responsible for the regression analysis and the construction of the empirical figures (Figures 6, 7, and 8). Jianan Xie was responsible for synthesizing the overall findings, organizing the presentation materials, and revising and editing the final written report.